B2B Travel Booking & Revenue Analytics

PHASE 5 - BUSINESS ANALYSIS WITH SQL

Complete Beginner-Friendly Documentation

Concepts | Business questions | SQL | Interpretation | Pitfalls | Portfolio thinking

Core Phase 5 idea: move from building trusted tables to using those tables to explain business performance. SQL is now used for decision-support questions, not only data preparation.

Project stack: PostgreSQL / pgAdmin | SQL | Python | Power BI | Excel

Learning mode: beginner-friendly, step-by-step

1. Phase 5 Overview

Earlier phases prepared the data: raw ingestion, profiling, cleaning, FX normalization and data marts. Phase 5 changes the focus. The main question is no longer “Is the table correct?” but “What is happening in the business, how large is the effect, and where should we investigate next?”

Pipeline position: RAW -> STAGING -> CLEAN -> FX NORMALIZATION -> DATA MARTS -> SQL BUSINESS ANALYSIS -> PYTHON -> POWER BI

The 8 lessons in Phase 5

Lesson	Business focus	Primary source
1	Overall business performance	analytics.fact_bookings_fx
2	Monthly growth and trends	analytics.monthly_growth
3	Travel agent performance	analytics.agent_performance
4	Destination performance	analytics.destination_performance
5	Supplier profitability	analytics.supplier_performance
6	Hotel performance	analytics.hotel_performance
7	Cancellation and refund analysis	analytics.cancellation_analysis
8	Turn query outputs into business insights	All Phase 5 outputs

Verified vs pending results

The overall KPI result from Lesson 1 was verified from the user’s pgAdmin screenshot. For several later lessons, the user chose “Next” without sending the query output. Those queries and concepts are complete, but specific top-agent, top-destination, top-supplier, top-hotel and cancellation-reason findings remain pending verification. This document does not invent those values.

Portfolio discipline: a query definition can be complete while a finding is still unverified. Never write an unobserved result as if it were a fact.

2. From SQL Output to Business Analysis

A good analyst does more than sort a table and report the first row. The same metric can mean different things depending on volume, profitability, cancellation behavior and sample size. Phase 5 repeatedly compares multiple KPIs together so that one number does not dominate the interpretation.

Three-part insight structure

```
FACT  ->  BUSINESS MEANING  ->  NEXT QUESTION / ACTION
```

Example: if GBV grows 15% but net revenue grows only 5%, the fact is the growth gap. The business meaning is that revenue efficiency did not grow at the same pace. The next step is to investigate supplier costs, discounting and booking mix. We should not claim one of those causes without evidence.

Important analytical distinctions

Do not confuse	Why it matters
GBV vs Net Revenue	GBV is booking value handled; revenue is what remains after supplier cost and discount.
Absolute revenue vs Margin %	A large supplier can generate more INR revenue but a lower percentage margin.
High rate vs reliable rate	A high cancellation rate from 2 bookings is less stable than the same rate from 500 bookings.
Correlation vs cause	Two metrics moving together does not prove one caused the other.
Ranking vs diagnosis	Top/bottom lists identify where to look; they do not by themselves explain why.

Analyst habit: report what the data shows, then label possible explanations as hypotheses to investigate.

3. Lesson 1 - Overall Business Performance

Business question: How is the analysis-ready B2B travel business performing overall?

KPIs

- Total bookings - business volume
- Total GBV in INR - total booking value handled
- Total net revenue in INR - value remaining after supplier cost and discount
- Average booking value - average GBV per booking
- Gross margin % - net revenue as a percentage of GBV
- Cancellation rate % - share of bookings marked Cancelled

```
SELECT
    COUNT(*) AS total_bookings,
    SUM(gross_booking_value_inr) AS total_gbv_inr,
    SUM(net_revenue_inr) AS total_net_revenue_inr,
    ROUND(SUM(gross_booking_value_inr) / NULLIF(COUNT(*), 0), 2)
        AS avg_booking_value_inr,
    ROUND((SUM(net_revenue_inr)
        / NULLIF(SUM(gross_booking_value_inr), 0)) * 100, 2)
        AS gross_margin_pct,
    ROUND((COUNT(*) FILTER (WHERE booking_status = 'Cancelled'))::numeric
        / NULLIF(COUNT(*), 0) * 100, 2)
        AS cancellation_rate_pct
FROM analytics.fact_bookings_fx
WHERE is_fx_missing = 0;
```

Verified project result

KPI	Verified value
Total Bookings	119,877
Total GBV	INR 26,151,677,378.28 (about INR 26.15B)
Total Net Revenue	INR 3,251,147,190.09 (about INR 3.25B)
Average Booking Value	INR 218,154.25
Gross Margin	12.43%
Cancellation Rate	11.64%

Reconciliation note: these values are mathematically consistent. INR 26.15B / 119,877 is about INR 218,154 per booking, and INR 3.25B / INR 26.15B is about 12.43%. This corrected the earlier Phase 3 transcription confusion.

4. How to Interpret the Overall KPIs

The overall KPI table is the executive baseline. It summarizes scale, monetization and cancellation exposure, but it is not a diagnosis by itself.

Metric-by-metric interpretation

Metric	What it tells us	What it does not tell us alone
119,877 bookings	Size of valid analysis population	Whether growth is accelerating or slowing
INR 26.15B GBV	Scale of booking value handled	Profitability
INR 3.25B net revenue	Absolute revenue generated	Revenue efficiency by segment
INR 218,154 ABV	Average value per booking	Whether value is concentrated in a few bookings
12.43% margin	Share of GBV retained as net revenue	Which costs or segments drive the margin
11.64% cancellation rate	Share of bookings cancelled	Root causes or refund severity

A concise professional summary is: The analysis-ready population contains 119,877 bookings, approximately INR 26.15B in GBV and INR 3.25B in net revenue. Overall gross margin is 12.43%, and 11.64% of bookings are cancelled.

Good reporting separates observation from explanation. “Cancellation rate is 11.64%” is an observation. “Cancellations are caused by supplier failures” would require additional evidence.

5. Lesson 2 - Monthly Growth and Trend Analysis

Business question: Is the business growing month by month, and which months show the strongest increases or declines? The Phase 4 mart analytics.monthly_growth already contains current-month and previous-month values.

Core growth formula

```
Growth % = (Current Month - Previous Month) / Previous Month x 100
```

```
SELECT
    booking_month,
    total_bookings,
    total_gbv_inr,
    total_net_revenue_inr,
    booking_growth_pct,
    gbv_growth_pct,
    net_revenue_growth_pct
FROM analytics.monthly_growth
ORDER BY booking_month;
```

Strongest and weakest GBV growth months

```
-- Highest growth
SELECT booking_month, total_gbv_inr, gbv_growth_pct
FROM analytics.monthly_growth
WHERE gbv_growth_pct IS NOT NULL
ORDER BY gbv_growth_pct DESC
LIMIT 5;

-- Largest decline
SELECT booking_month, total_gbv_inr, gbv_growth_pct
FROM analytics.monthly_growth
WHERE gbv_growth_pct IS NOT NULL
ORDER BY gbv_growth_pct ASC
LIMIT 5;
```

Compare GBV growth with revenue growth

```
SELECT
    booking_month,
    gbv_growth_pct,
    net_revenue_growth_pct
FROM analytics.monthly_growth
WHERE gbv_growth_pct IS NOT NULL
ORDER BY booking_month;
```

Interpretation: GBV growth and net revenue growth can diverge. That divergence is a signal to investigate costs, discounting or booking mix; it is not proof of any one cause.

Project status: highest and lowest growth months were not reported back during the lesson, so those exact findings remain pending.

6. Lesson 3 - Travel Agent Performance Analysis

Business question: Which travel agents drive booking volume and net revenue, and which agents show high cancellation rates?

Top by booking volume

```
SELECT
    agent_id, total_bookings, total_gbv_inr,
    total_net_revenue_inr, cancellation_rate_pct
FROM analytics.agent_performance
ORDER BY total_bookings DESC
LIMIT 10;
```

Top by net revenue

```
SELECT
    agent_id, total_bookings, total_gbv_inr,
    total_net_revenue_inr, gross_margin_pct,
    cancellation_rate_pct
FROM analytics.agent_performance
ORDER BY total_net_revenue_inr DESC
LIMIT 10;
```

High-cancellation agents with a minimum sample

```
SELECT
    agent_id, total_bookings, cancelled_bookings,
    cancellation_rate_pct, total_gbv_inr,
    total_net_revenue_inr
FROM analytics.agent_performance
WHERE total_bookings >= 20
ORDER BY cancellation_rate_pct DESC
LIMIT 10;
```

The >= 20 filter is not a universal law. It is a practical threshold used here to reduce noisy comparisons from tiny samples. In a production project, the threshold should be justified using business volume or statistical stability.

Key lesson: “top by bookings” and “top by revenue” may be different agents. Volume, monetization and cancellation quality should be viewed together.

Project status: specific top-agent outputs were not submitted, so no agent ID is presented as a verified finding.

7. Lesson 4 - Destination Performance Analysis

Business question: Which destinations generate the most booking value and net revenue, and which destinations have higher cancellation rates?

```
-- Top destinations by GBV
SELECT
    destination_id, total_bookings, total_gbv_inr,
    total_net_revenue_inr, gross_margin_pct,
    cancellation_rate_pct
FROM analytics.destination_performance
ORDER BY total_gbv_inr DESC
LIMIT 10;

-- Top destinations by net revenue
SELECT
    destination_id, total_bookings, total_gbv_inr,
    total_net_revenue_inr, gross_margin_pct,
    cancellation_rate_pct
FROM analytics.destination_performance
ORDER BY total_net_revenue_inr DESC
LIMIT 10;
```

High-cancellation destinations

```
SELECT
    destination_id, total_bookings, cancelled_bookings,
    cancellation_rate_pct, total_gbv_inr,
    total_net_revenue_inr
FROM analytics.destination_performance
WHERE total_bookings >= 100
ORDER BY cancellation_rate_pct DESC
LIMIT 10;
```

A destination can be commercially important because of high GBV, yet still deserve operational attention if its cancellation rate is high or its margin is weak. This is why a multi-KPI view is more useful than a one-column ranking.

Threshold note: >= 100 bookings was used to focus on destinations with meaningful activity. The choice is analytical, not a fixed industry rule.

Project status: exact destination winners / high-cancellation destinations remain pending because lesson outputs were skipped.

8. Lesson 5 - Supplier Profitability Analysis

Business question: Which suppliers generate scale, which generate net revenue, and how do their margins differ? Supplier analysis is especially important because supplier cost directly enters the net revenue formula.

Net Revenue = GBV - Supplier Cost - Discount

Top suppliers by GBV and net revenue

```
SELECT
    supplier_id, total_bookings, total_gbv_inr,
    total_supplier_cost_inr, total_net_revenue_inr,
    gross_margin_pct, cancellation_rate_pct
FROM analytics.supplier_performance
ORDER BY total_gbv_inr DESC
LIMIT 10;

SELECT
    supplier_id, total_bookings, total_gbv_inr,
    total_supplier_cost_inr, total_net_revenue_inr,
    gross_margin_pct, cancellation_rate_pct
FROM analytics.supplier_performance
ORDER BY total_net_revenue_inr DESC
LIMIT 10;
```

Compare stronger and weaker margins

```
-- Higher margins
SELECT supplier_id, total_bookings, total_gbv_inr,
       total_net_revenue_inr, gross_margin_pct
FROM analytics.supplier_performance
WHERE total_bookings >= 100
ORDER BY gross_margin_pct DESC
LIMIT 10;

-- Lower margins
SELECT supplier_id, total_bookings, total_gbv_inr,
       total_supplier_cost_inr, total_net_revenue_inr,
       gross_margin_pct
FROM analytics.supplier_performance
WHERE total_bookings >= 100
ORDER BY gross_margin_pct ASC
LIMIT 10;
```

Important: a low-margin supplier is not automatically a “bad” supplier. It may serve strategically important destinations, have different contract economics or carry a different booking mix. The SQL identifies where to investigate.

Data-quality checkpoint: Phase 4 produced 161 supplier buckets including UNKNOWN, but the exact UNKNOWN booking count needs a final diagnostic because a previously reported value of 1 did not reconcile with the earlier missing-supplier profile. Do not use that count as a published finding yet.

9. Lesson 6 - Hotel Performance Analysis

Business question: Which hotels receive the most booking volume and net revenue, and which hotels show high cancellation rates or unusual margins?

```
-- Highest booking volume
SELECT
    hotel_id, total_bookings, total_gbv_inr,
    total_net_revenue_inr, gross_margin_pct,
    cancellation_rate_pct
FROM analytics.hotel_performance
ORDER BY total_bookings DESC
LIMIT 10;

-- Highest net revenue
SELECT
    hotel_id, total_bookings, total_gbv_inr,
    total_net_revenue_inr, gross_margin_pct,
    cancellation_rate_pct
FROM analytics.hotel_performance
ORDER BY total_net_revenue_inr DESC
LIMIT 10;
```

High cancellations and margin comparisons

```
-- High cancellation rate with meaningful activity
SELECT hotel_id, total_bookings, cancelled_bookings,
       cancellation_rate_pct, total_gbv_inr,
       total_net_revenue_inr
FROM analytics.hotel_performance
WHERE total_bookings >= 50
ORDER BY cancellation_rate_pct DESC
LIMIT 10;

-- Higher margin hotels
SELECT hotel_id, total_bookings, total_gbv_inr,
       total_net_revenue_inr, gross_margin_pct
FROM analytics.hotel_performance
WHERE total_bookings >= 50
ORDER BY gross_margin_pct DESC
LIMIT 10;
```

The >= 50 threshold protects the comparison from very small hotel samples. High-volume, low-margin hotels and lower-volume, high-margin hotels can both matter; the correct business response depends on strategy and contract context.

Project status: exact hotel IDs for top volume, revenue, cancellation and margin were not reported, so those findings remain pending verification.

10. Lesson 7 - Cancellation and Refund Analysis

Business question: What are the main cancellation reasons, and what financial impact is associated with refunds and cancellation fees?

Cancellation reason summary

```
SELECT
    cancellation_reason,
    total_cancellations,
    total_refund_amount_inr,
    total_cancellation_fee_inr,
    avg_refund_amount_inr,
    avg_cancellation_fee_inr
FROM analytics.cancellation_analysis
ORDER BY total_cancellations DESC;
```

Highest total refund and highest average refund

```
SELECT cancellation_reason, total_cancellations,
       total_refund_amount_inr, avg_refund_amount_inr
FROM analytics.cancellation_analysis
ORDER BY total_refund_amount_inr DESC;

SELECT cancellation_reason, total_cancellations,
       total_refund_amount_inr, avg_refund_amount_inr
FROM analytics.cancellation_analysis
ORDER BY avg_refund_amount_inr DESC;
```

Fee-to-refund ratio

```
SELECT
    cancellation_reason,
    total_refund_amount_inr,
    total_cancellation_fee_inr,
    ROUND(total_cancellation_fee_inr
        / NULLIF(total_refund_amount_inr, 0) * 100, 2)
        AS fee_to_refund_pct
FROM analytics.cancellation_analysis
ORDER BY fee_to_refund_pct DESC;
```

Interpretation: high total refund can come from many cancellations, high-value cancellations, or both. Total refund and average refund answer different questions.

Verified Phase 4 structure: cancellation_analysis has 8 unique reasons, 0 duplicate reasons, and cancellation_difference = 0. Specific reason names and refund rankings were not submitted during Phase 5.

11. Lesson 8 - Turn SQL Results into Business Insights

The final analytical step is to communicate a business story without overstating what the data proves. A useful insight connects an observed fact to its commercial relevance and then recommends a focused investigation or action.

Example insight patterns

Observed pattern	Useful interpretation	Possible next analysis
GBV rises faster than revenue	Booking value grew, but revenue efficiency lagged	Compare supplier cost, discount and segment mix
Agent has high volume and high cancellations	Agent is important but cancellation behavior may create operational / refund exposure	Compare reason mix, destination mix and booking lead time
Destination has high GBV but low margin	Commercial scale does not translate proportionally into revenue	Check supplier economics and discounts
Supplier has strong absolute revenue but lower margin	Scale is high but unit economics are thinner	Review cost rate / contract mix
Cancellation reason has high total refunds	Reason creates material refund exposure	Separate volume effect from average refund severity

What not to do

- Do not call a supplier, agent, hotel or destination “best” or “worst” from one KPI alone.
- Do not claim a cause when the query only shows an association.
- Do not compare rates from tiny samples without a minimum-volume check.
- Do not mix local currencies after Phase 3; use the INR-normalized analytics layer.
- Do not join payment transactions directly to booking facts and then sum GBV; that can duplicate booking value.

Professional language: “This pattern suggests further investigation” is often more accurate than “This proves the cause.”

12. SQL Concepts Reinforced in Phase 5

Concept	Purpose in Phase 5
COUNT(*)	Measure booking / cancellation volume
SUM()	Aggregate GBV, revenue, supplier cost, refunds and fees
FILTER (WHERE ...)	Conditional counts such as cancellations
NULLIF()	Prevent divide-by-zero errors in rate calculations
ROUND()	Present business KPIs at practical precision
ORDER BY ... DESC / ASC	Identify high and low metric values
LIMIT	Create focused top-N / bottom-N views
WHERE total_bookings >= N	Apply minimum-volume thresholds
Prebuilt marts	Reuse controlled business definitions rather than rewriting logic

Why marts matter here

Phase 5 queries are short because Phase 4 already created the right grain. That is a sign of good analytics architecture. Complex preparation belongs upstream; business queries should be readable enough that another analyst can understand the question quickly.

Good data engineering makes business analysis simpler, safer and more repeatable.

13. Phase 5 Validation and Completion Checklist

Before publishing Phase 5 findings in GitHub, a dashboard or an interview case study, rerun the pending outputs and record the exact results.

Checkpoint	Current status
Overall KPI baseline	VERIFIED: 119,877 bookings; INR 26.15B GBV; INR 3.25B net revenue; INR 218,154.25 ABV; 12.43% margin; 11.64% cancellation
Highest / lowest monthly GBV growth	PENDING output capture
Top agent by bookings / revenue	PENDING output capture
Highest cancellation agent >= 20 bookings	PENDING output capture
Top destination by GBV / revenue	PENDING output capture
Highest cancellation destination >= 100 bookings	PENDING output capture
Top supplier by GBV / revenue / margin	PENDING output capture
Supplier UNKNOWN bucket diagnostic	PENDING reconciliation
Top hotel by volume / revenue / margin	PENDING output capture
High-cancellation hotel >= 50 bookings	PENDING output capture
Most common cancellation reason / highest refund	PENDING output capture

Phase 5 SQL learning is complete. The pending items are result-capture checkpoints, not missing concepts. They should be completed before writing final project recommendations.

14. Portfolio and Interview Talking Points

How to explain Phase 5 in an interview

A concise explanation: “After building the clean and FX-normalized analytics layer, I created business marts at monthly, agent, destination, supplier, hotel and cancellation-reason grain. I then used SQL to measure scale, growth, revenue, margins and cancellation behavior. I applied minimum-volume thresholds to reduce noisy rate comparisons and separated observed patterns from possible causes.”

Likely interview questions

- Why is GBV different from net revenue?
- Why did you calculate margin from aggregated revenue and GBV instead of averaging row-level margin?
- Why did you use minimum booking thresholds for cancellation-rate comparisons?
- Why can the top agent by bookings differ from the top agent by net revenue?
- What does it mean if GBV grows faster than net revenue?
- Why is a high refund total not necessarily the same as a high average refund?
- How would you investigate a low-margin supplier?
- How did you protect the analysis from multi-currency errors?
- How did you avoid payment joins duplicating booking value?

Strong answer principles

- Start with the business definition of the metric.
- State the grain of the table being analyzed.
- Explain the SQL choice in plain language.
- Mention validation / reconciliation when relevant.
- Avoid claiming causality unless the analysis actually establishes it.

15. Phase 5 Final Summary and Phase 6 Handoff

Phase 5 completed the SQL business-analysis layer of the portfolio project. You now know how to use trusted marts to answer executive, growth, partner, market, profitability and cancellation questions.

What you can now do

- Calculate and interpret overall business KPIs.
- Analyze month-over-month booking, GBV and revenue growth.
- Compare agents on volume, revenue and cancellation behavior.
- Compare destinations on commercial scale and cancellation rates.
- Analyze supplier cost, net revenue and margin together.
- Analyze hotel demand, profitability and cancellation behavior.
- Separate cancellation frequency, refund total, average refund and cancellation fees.
- Use thresholds to make percentage comparisons more meaningful.
- Write business insights that separate facts, interpretations and hypotheses.

Next phase: Phase 6 - Python EDA and Root-Cause Analysis. Python will be used to inspect distributions, relationships, segment patterns and root-cause candidates more flexibly than fixed SQL summaries.

Project pipeline after Phase 5

```
RAW DATA
   -> STAGING
   -> CLEAN DATA
   -> FX NORMALIZATION
   -> ANALYTICS MARTS
   -> SQL BUSINESS ANALYSIS   [PHASE 5]
   -> PYTHON EDA / ROOT-CAUSE ANALYSIS
   -> POWER BI
   -> FINAL BUSINESS INSIGHTS
```

Final reminder: the verified executive baseline is ready to use. Capture the pending top-N query outputs before publishing specific partner, destination, hotel or cancellation-reason claims.